CM3121 – Advanced Organic Chemistry

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Enolates and α-alkylation reactions

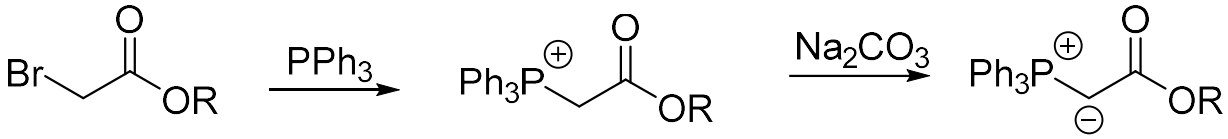
Readings: Clayden 2nd ed, Chapter 25 and McMurry, 10th ed, Chapter 22

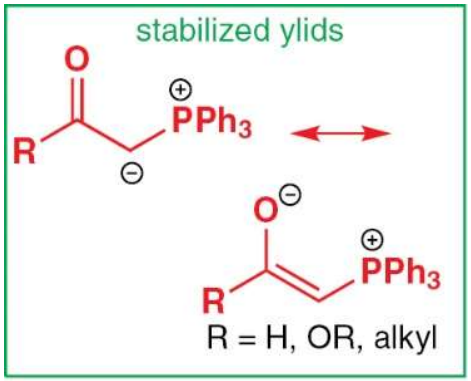
Outlines

1. Enolate formation
2. α- Alkylation reactions
3. 1,3-Dicarbonyl compounds
4. Enolate from other functional groups
5. Asymmetric alkylation reaction

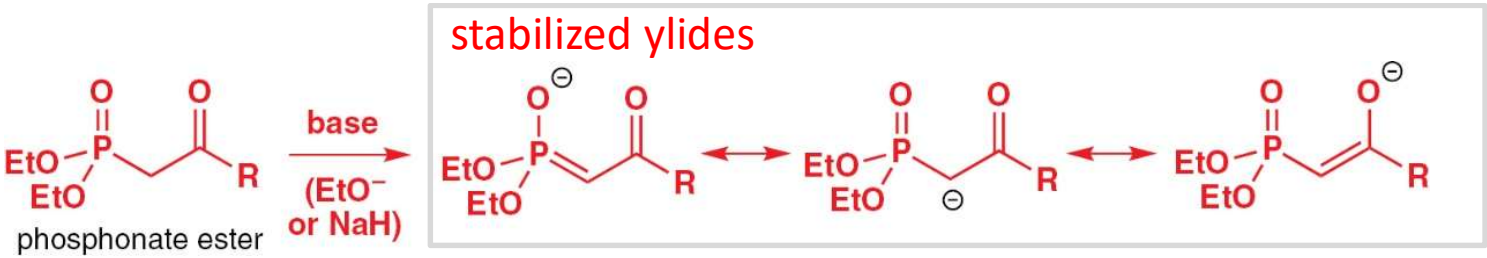
Ylides for Wittig reactions (from previous lecture)

Those with conjugating or anion stabilizing substituents adjacent to the negative charge (stabilized ylides)





Horner-Wadsworth-Emmons (HWE) reactions with stabilized ylides.

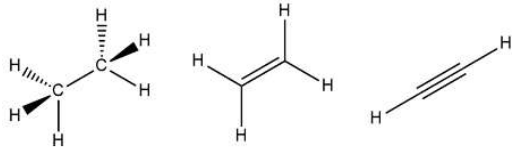


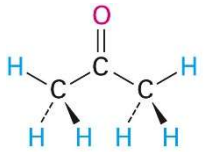
I. Enolate formation

pK_a Table

The pK_a values for the α-hydrogens for most simple aldehydes or ketones are of the order of 19-20

Ethyne pKa = 25
Ethylene pKa = 44
Ethane pKa = 50





Acetone
(pK_a = 19.3)

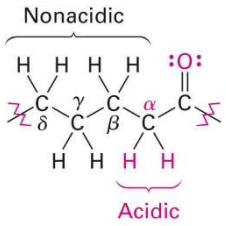
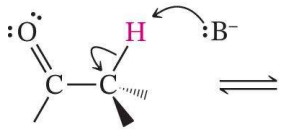


Table 22.1 Acidity Constants for Some Organic Compounds

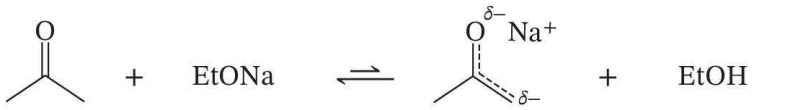
Functional group	Example	pK_a
Carboxylic acid	CH_3COH	5
1,3-Diketone	$CH_3CCH_2CCH_3$	9
3-Keto ester	$CH_3CCH_2COCH_3$	11
1,3-Diester	$CH_3OCCH_2COCH_3$	13
Alcohol	CH_3OH	16
Acid chloride	CH_3CCl	16
Aldehyde	CH_3CH	17
Ketone	CH_3CCH_3	19
Thioester	CH_3CSCH_3	21
Ester	CH_3COCH_3	25
Nitrile	$CH_3C{\equiv}N$	25
N,N-Dialkylamide	$CH_3CN(CH_3)_2$	30
Dialkylamine	$HN(i\text{-}C_3H_7)_2$	36

The acidity of the α hydrogen of carbonyl groups

When a carbonyl compound loses an α hydrogen, the anion formed is called enolate. Enolate ion is stabilized by resonance (electron delocalization)



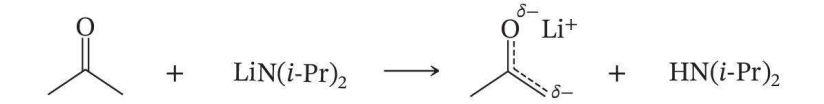
How much enolate form depends on the strength of the base used



Weaker acid ($pK_a = 19$) **Weaker base** **Stronger base** **Stronger acid** ($pK_a = 16$)

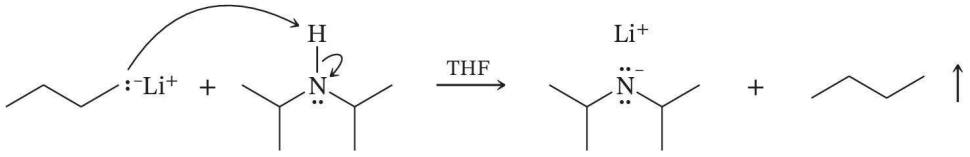
LDA base

How much enolate form depends on the strength of the base used
Strong base: LDA (lithium diisopropylamide)



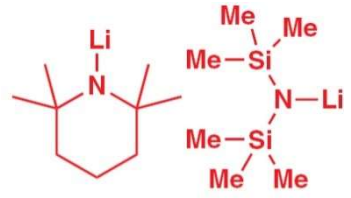
Ketone (stronger acid) ($pK_a = 19$) | **LDA (stronger base)** | **Enolate (weaker base)** | **Diisopropylamine (weaker acid) ($pK_a = 38$)**

LDA is prepared from butyllithium (BuLi) and diisopropylamine.
LDA is a strong base that is NOT nucleophilic



Butyllithium (BuLi) | **Diisopropylamine ($pK_a = 38$)** | **Lithium diisopropylamide [LDA or LiN(*i*-Pr)$_2$]** | **Butane ($pK_a = 50$)**

Other strong bases that are NOT nucleophilic

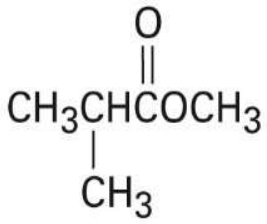


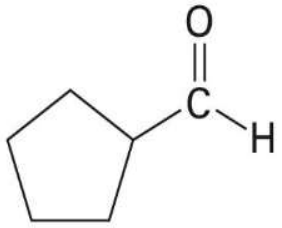
LTMP or LiTMP

LHMDS

Enolates

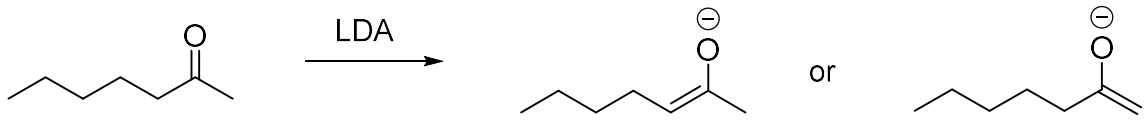
Provide the structure of the enolate in the reaction of LDA and the following compounds





Enolate ion formation

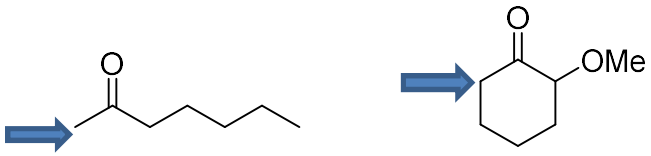
For an unsymmetrical ketone, 2 possible enolates can formed.



The thermodynamic enolate: most stable among the possible enolates
The kinetic enolate: the one that is formed fastest (C-H is more accessibly)

Which enolate predominates in the reaction depends on the reaction conditions.

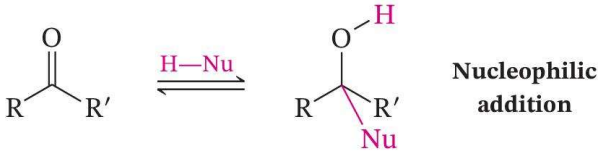
For kinetic control (typically at low temperature)

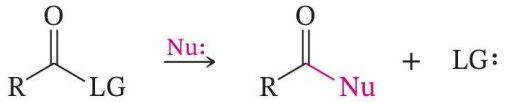


100: 0
LDA -78 °C

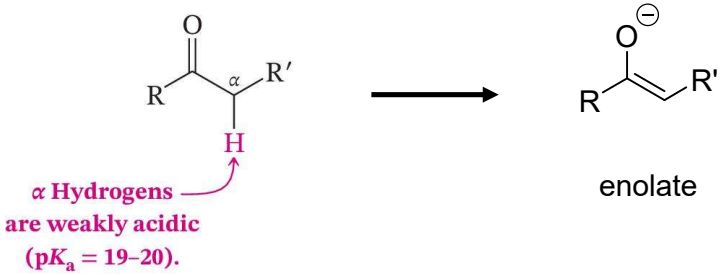
85:15
LDA, -78 °C

Carbonyl compounds as electrophiles





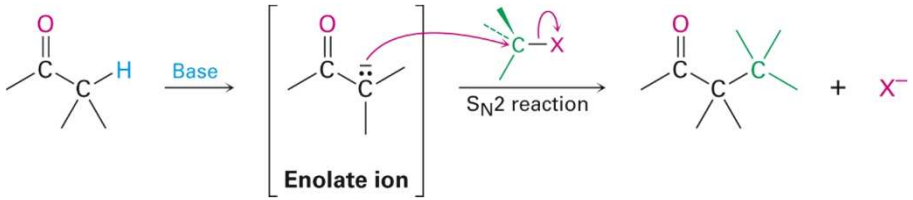
Deprotonation of carbonyl compounds give enolates → excellent nucleophiles



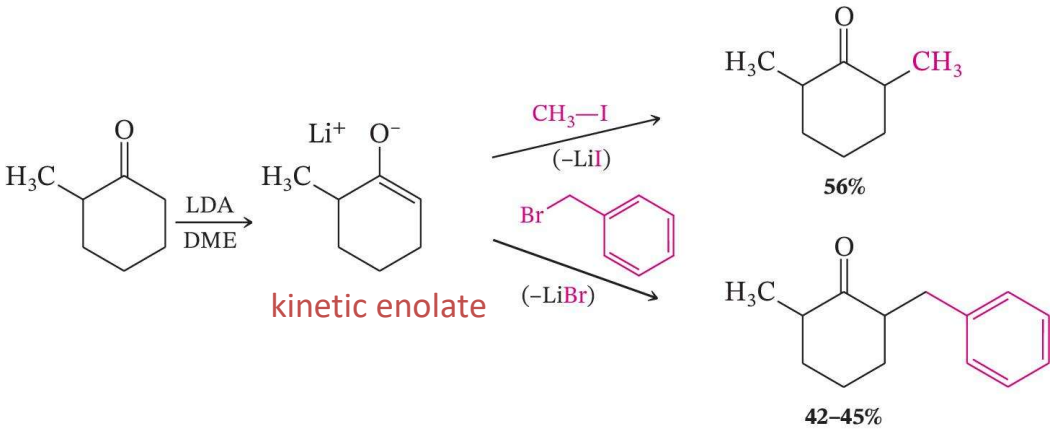
The carbon atom of an enolate ion is electron-rich and highly reactive toward electrophiles (E^+)

II. α-Alkylation reactions

Reaction between an enolate and an alkyl halide to produce an α- substituted carbonyl compound.



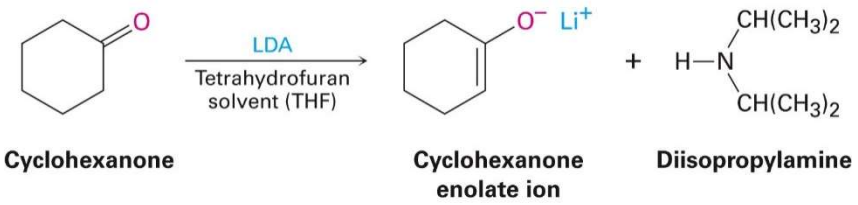
Examples:



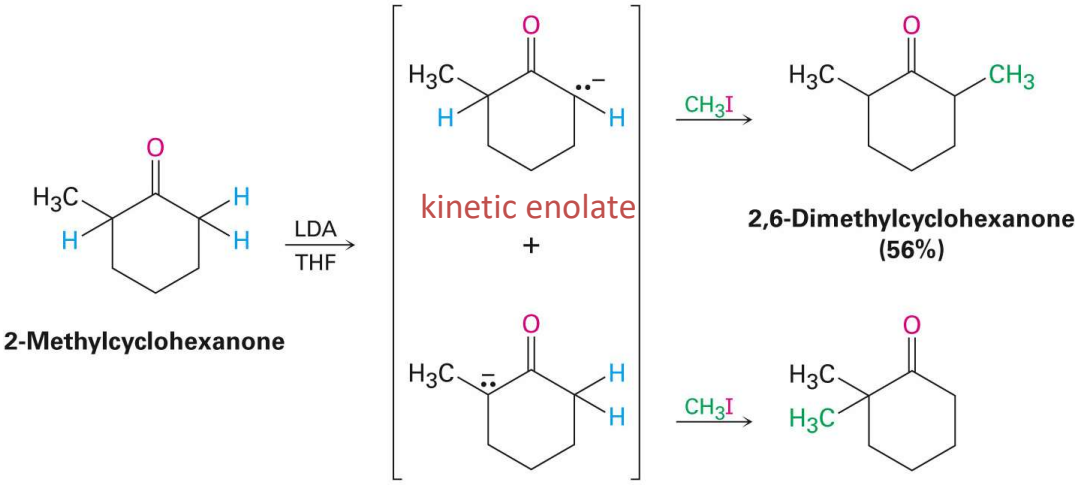
S_N2 reaction:

- The leaving group X can be chloride, bromide, iodide, or tosylate
- R should be primary or methyl and preferably should be allylic or benzylic
- Secondary halides react poorly, and tertiary halides don't react at all because of competing elimination

Enolate ion formation



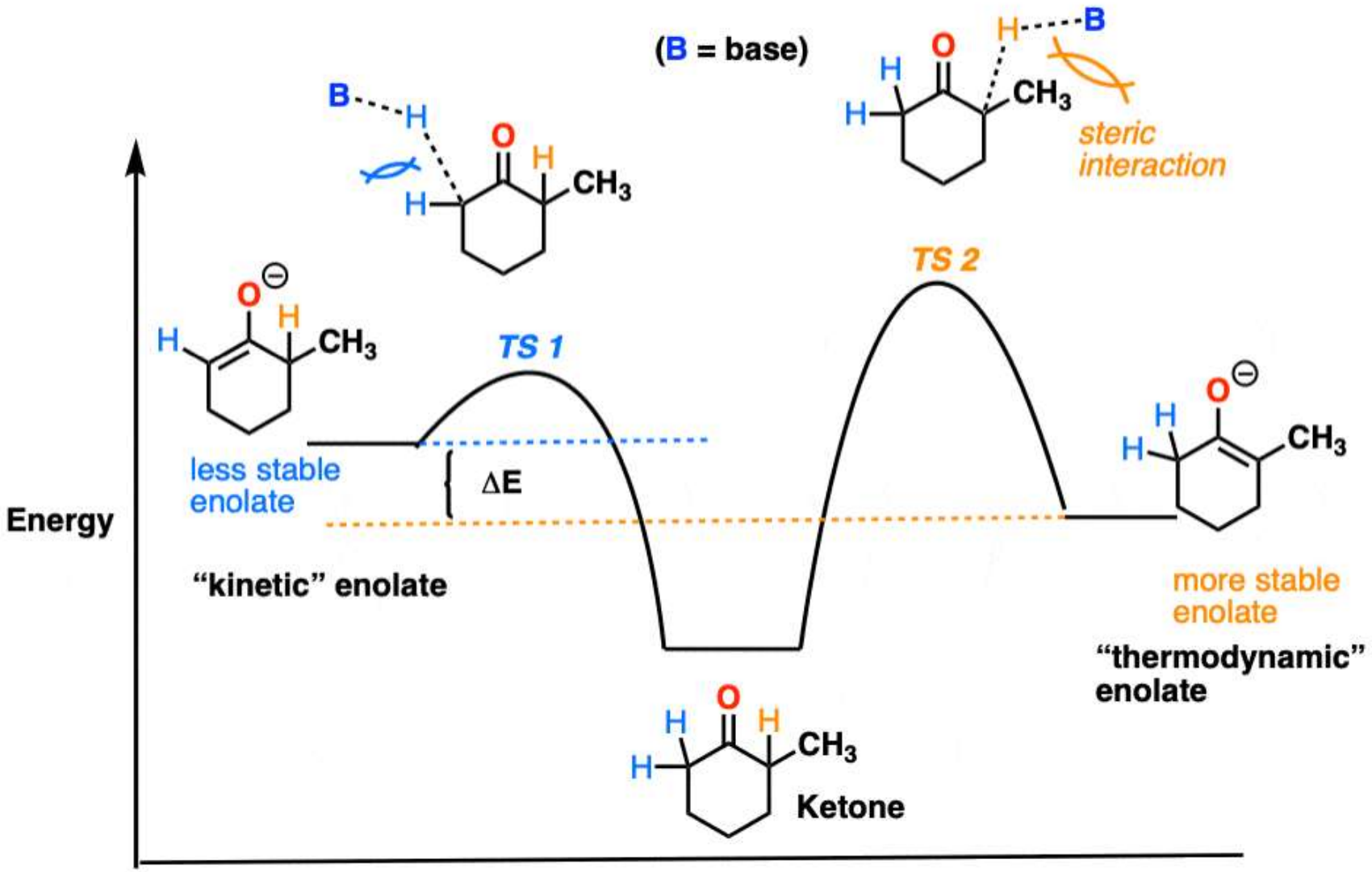
Cyclohexanone | **Cyclohexanone enolate ion** | **Diisopropylamine**



2-Methylcyclohexanone

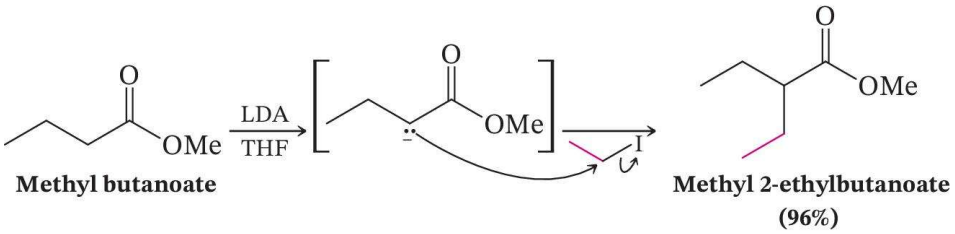
2,6-Dimethylcyclohexanone (56%)

2,2-Dimethylcyclohexanone (6%)

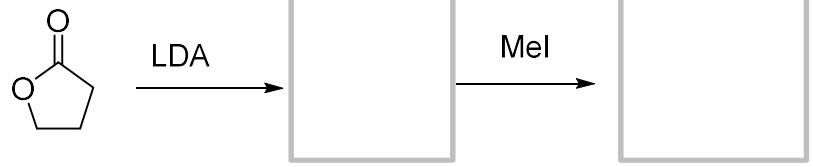

(B = base)
B
H
O
H
CH3
H
steric
interaction
TS 2
TS 1
less stable
enolate
ΔE
Energy
"kinetic" enolate
more stable
enolate
"thermodynamic"
enolate
Ketone

Alkylation reaction of esters

Examples



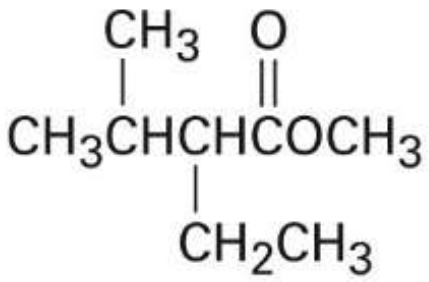
Methyl butanoate **Methyl 2-ethylbutanoate (96%)**



butyrolactone 88%

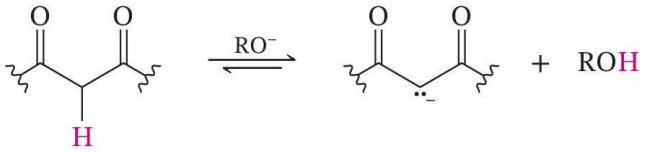
Exercise

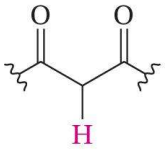
Synthesize the following compound by alpha-substitution reaction of enolate.



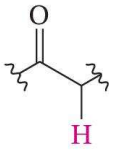
III. 1,3 - Dicarbonyl compounds

Hydrogen atoms that are between 2 carbonyl groups (β-dicarbonyl compounds, β-ketoesters) have pKa values in the range of 9-11. A much weaker base than LDA, such as alkoxide, can be used to form an enolate



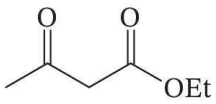


pK_a 9–11

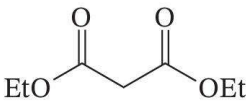


pK_a 18–20

2 compounds have a broad use in organic synthesis



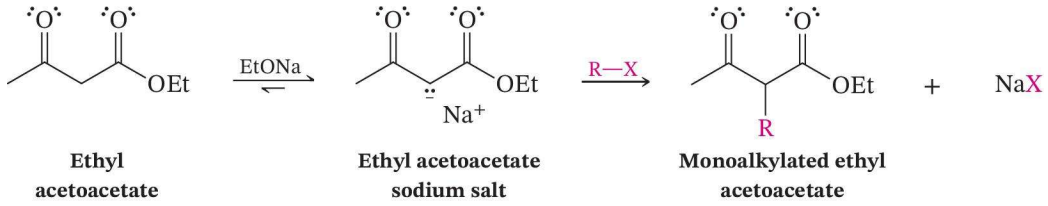
Acetoacetic ester
(ethyl acetoacetate;
ethyl 3-oxobutanoate)



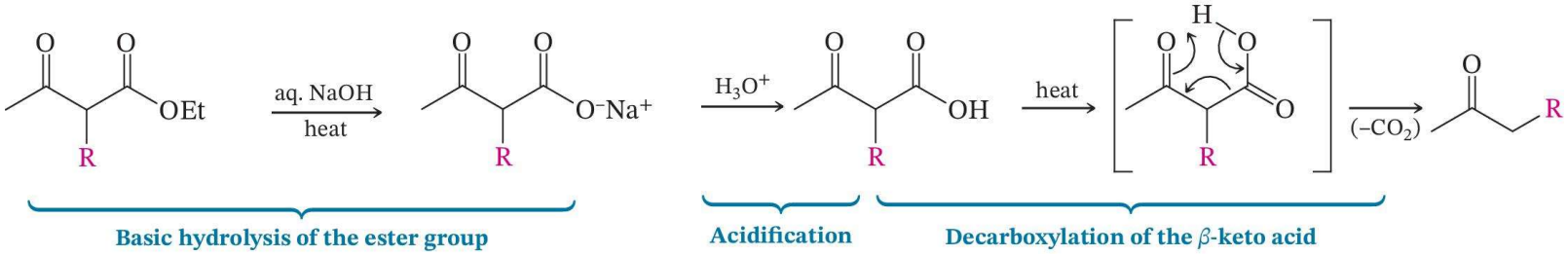
Diethyl malonate
(diethyl 1,3-propanedicarboxylic acid)

Alkylation with 1,3 - dicarbonyl compounds

3.1 Acetoacetic esters synthesis: Ethyl acetoacetate can be easily converted to an enolate using sodium ethoxide. The resulting enolate can undergo alkylation reaction.

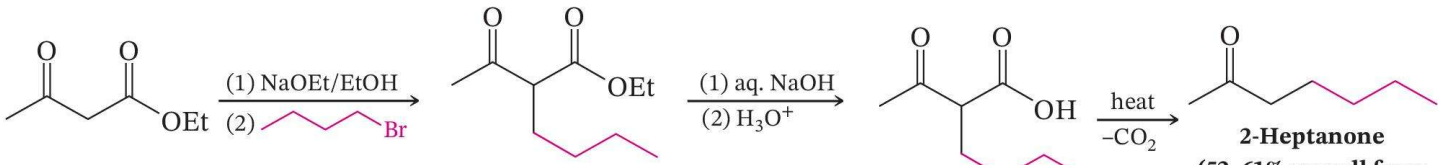


The product can be hydrolyzed and subsequently undergo decarboxylation reaction



Alkylation with dicarbonyl compounds

Acetoacetic esters synthesis:

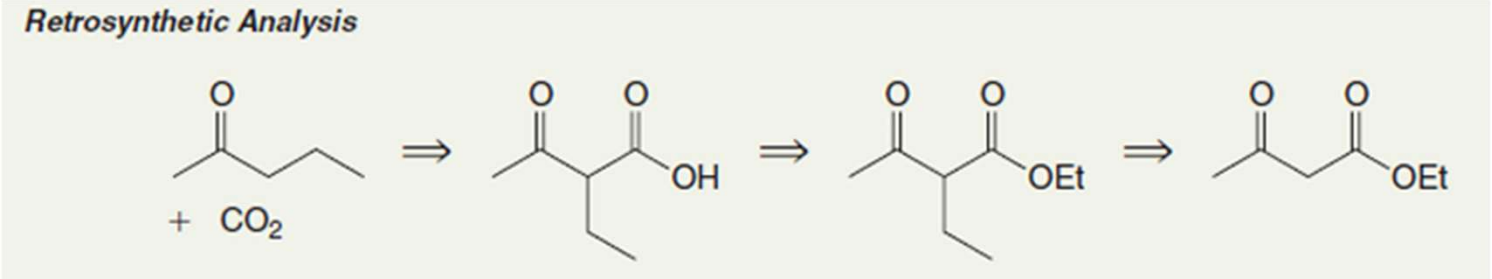


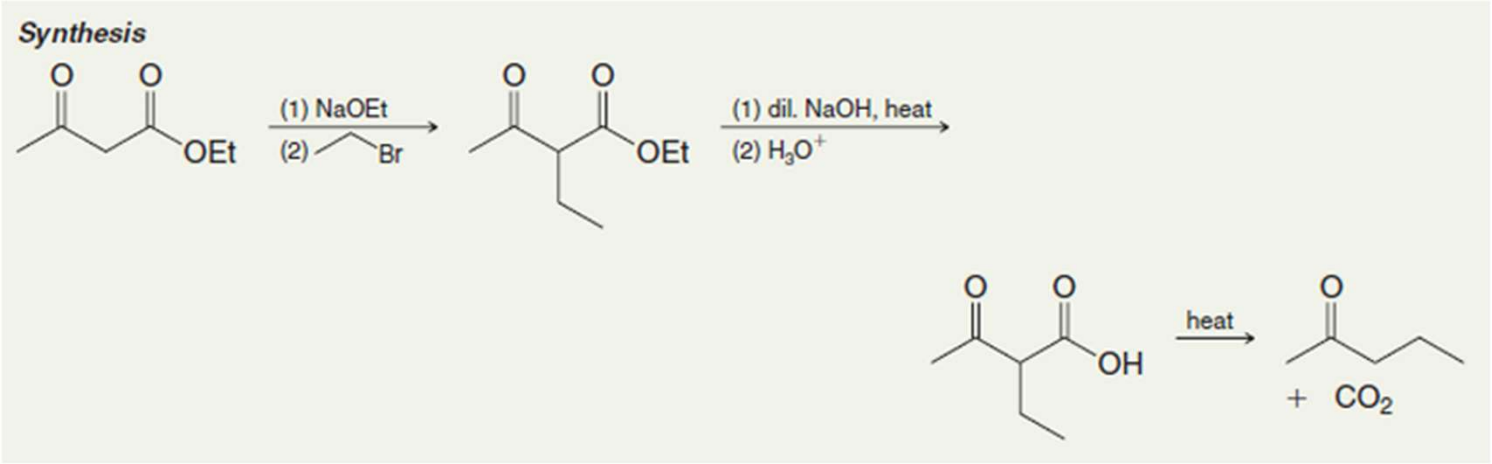
Ethyl acetoacetate (acetoacetic ester)

Ethyl butylacetoacetate (69–72%)

2-Heptanone (52–61% overall from ethyl acetoacetate)

Retrosynthetic analysis

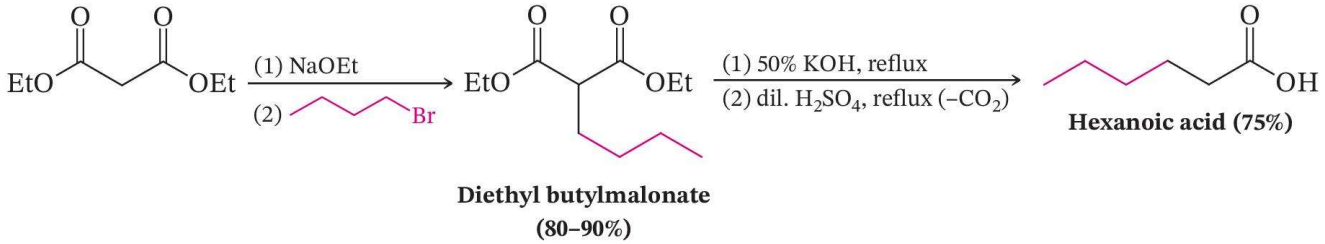




Alkylation with 1,3 - dicarbonyl compounds

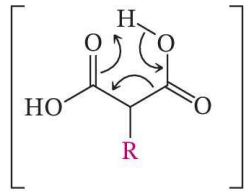
3.2 Malonic ester synthesis: diethyl malonate can be easily converted to an enolate using sodium ethoxide. The resulting enolate can undergo alkylation reaction.

A Malonic Ester Synthesis of Hexanoic Acid



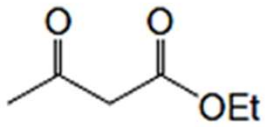
Diethyl butylmalonate (80–90%)

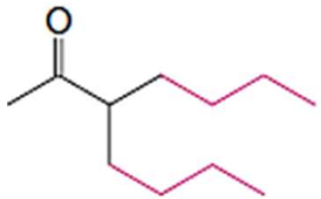
Hexanoic acid (75%)



Exercise

Synthesis the target molecule starting from ethyl acetoacetate .





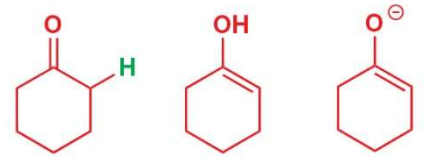
3-Butyl-2-heptanone

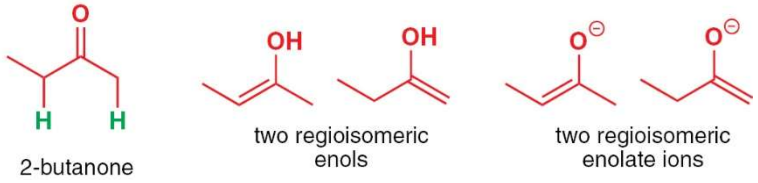
Target molecule

IV. Enolate from various functional groups

Ketones

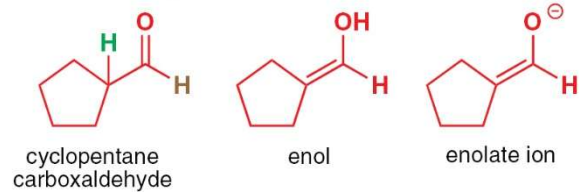
Enolizable ketones



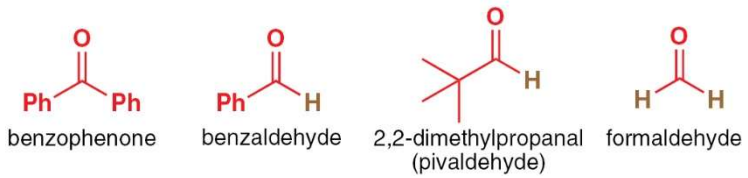


Aldehydes

Enolizable aldehyde



Non-enolizable carbonyl compounds

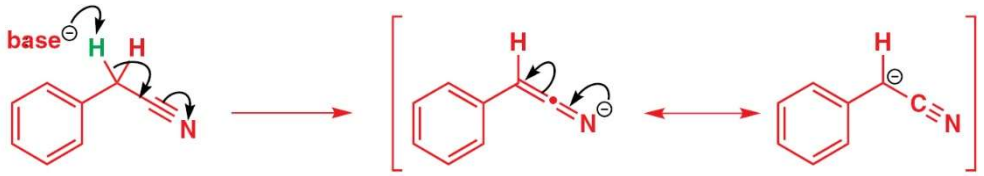


Esters



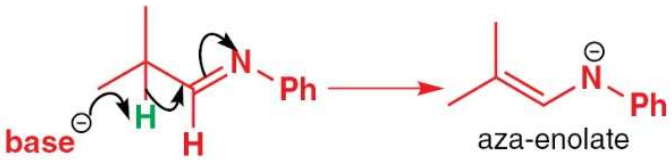
Enolate from other functional groups (nitrogen containing)

Nitriles

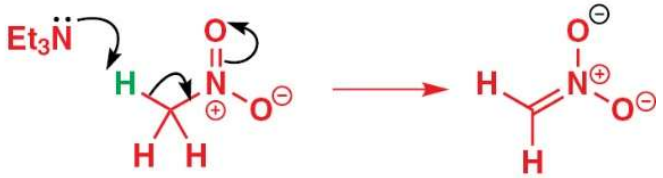


Imines (C=N) and nitro compounds (N=O)

Formation of aza-enolate



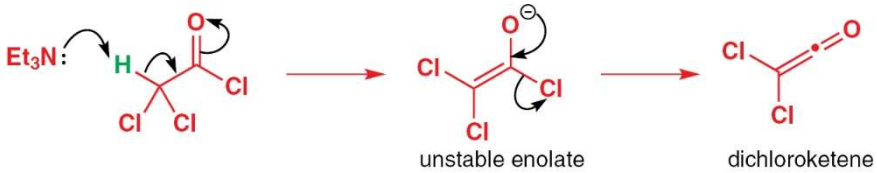
Formation of nitromethane anion



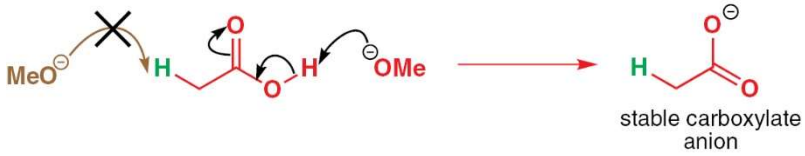
Requirements: an electron-withdrawing functional group with one π bond joined to a saturated carbon atom having at least one hydrogen atom

Functional groups that do not form enolates under typical conditions

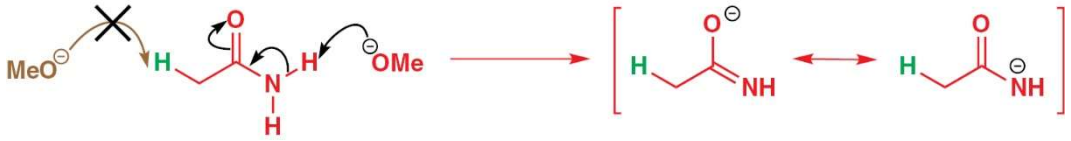
Acid chlorides (NO enolate formation)



Carboxylic acids (NO enolate formation)

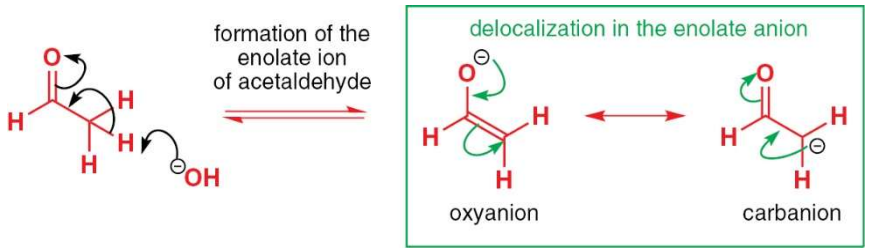


Amides with NH (NO enolate formation)

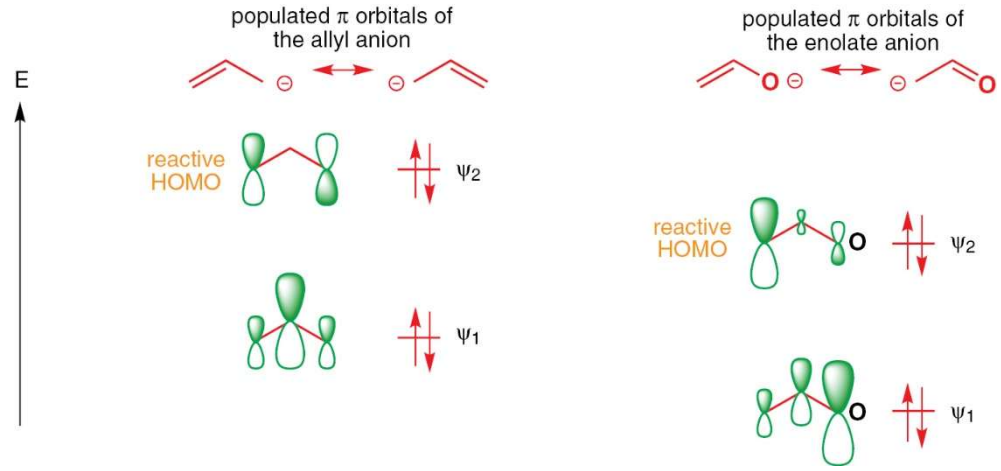


V. Enolate reactivity: C vs O

Enolate is a 3 atom 4 electrons system.



Molecular orbitals:



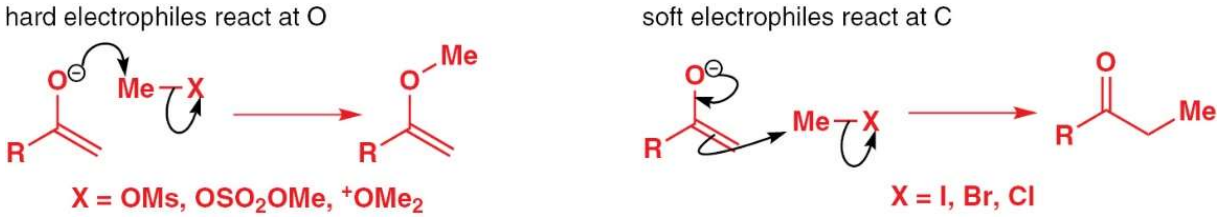
HOMO has a larger orbital on the terminal carbon

Enolate reactivity: C vs O

1. Electrophile:

soft electrophiles: reaction at C

hard electrophiles: reaction at O

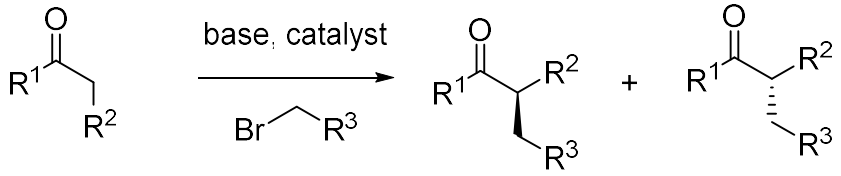


2. Metal cation:

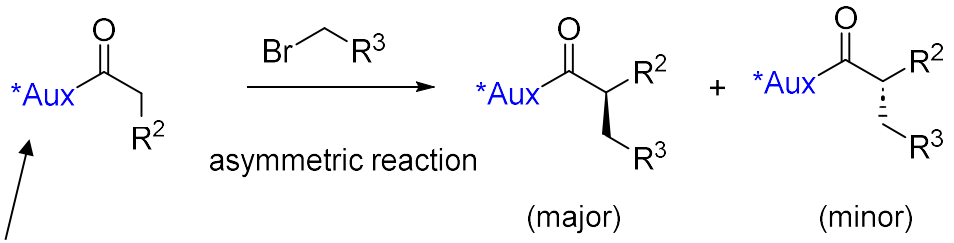
Larger alkali metals (Cs > K> Na> Li) give more separate ion pairs → react at O

V. Asymmetric alkylation reaction

Normal alkylation reaction give racemic mixtures



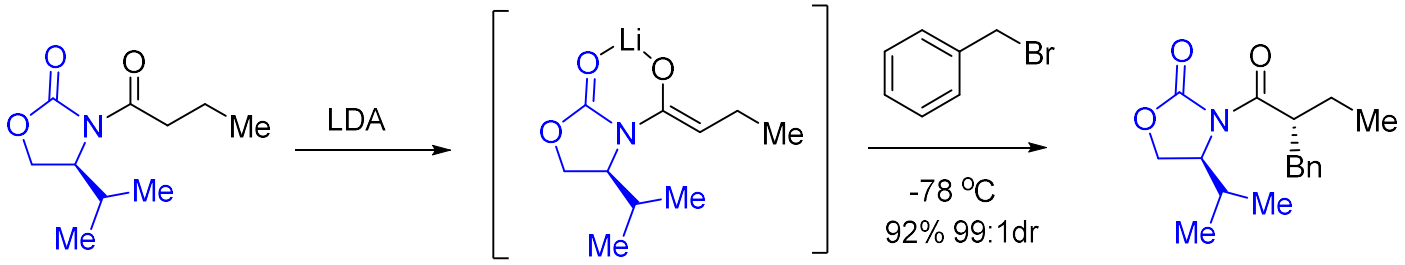
If a chiral group presents near the reacting center (Aux), it can influence the stereomeric distribution



This group is called chiral auxiliary
(it must contain chiral center(s))

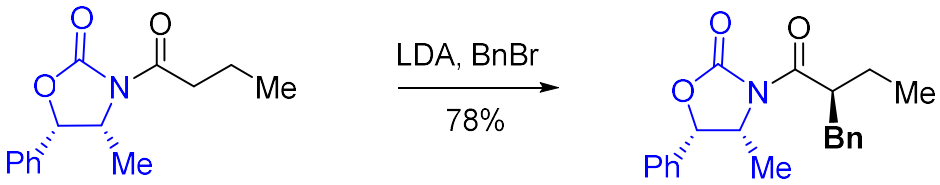
Asymmetric alkylation reaction

Evans chiral auxiliaries: oxazolidinones



Z- enolate

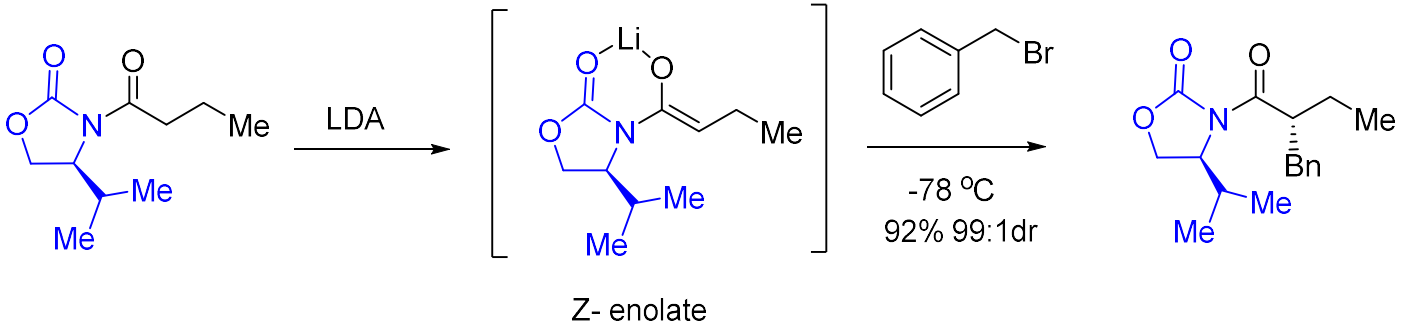
To synthesize the other stereoisomer:



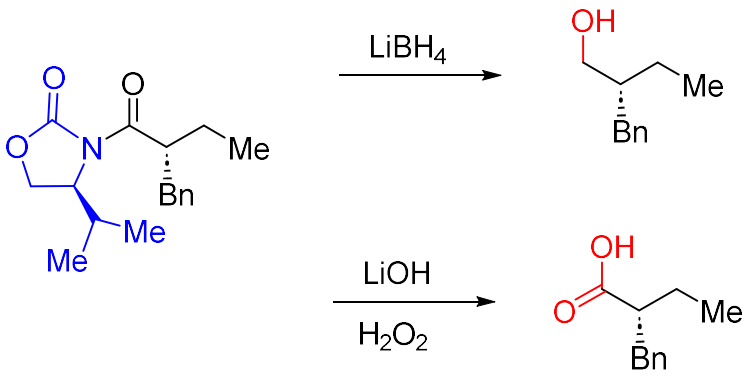
99:1 dr

Evans, J. Am. Chem. Soc. 1982, 104, 1737-1739

Remove the chiral auxiliary from the products



Substrate scope: the Evans asymmetric alkylation reaction works best with excellent S_N2 electrophiles (allyl, benzyl, methyl, X-CH_2OR, etc).



Summary

- Formation of α-enolates of ketones and esters using strong bases
- The formation of “kinetic” enolates (less substituted C=C) using bulky bases
- Alkylation reactions of α-enolates through nucleophilic substitution
- Alkylation reactions of 1,3 - dicarbonyl compounds
  - ethyl acetoacetate: adding -$CH_2(C=O)CH_3$ upon hydrolysis and decarboxylation
  - Diethyl malonate: adding -CH_2COOH upon hydrolysis and decarboxylation
- Enolate from other functional groups